SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID –** 25041
- **Problem Statement Title-** AI-Driven Unified Data Platform for Oceanographic, Fisheries, and Molecular Biodiversity Insights
- **Theme-** Renewable / Sustainable Energy
- **PS Category-** Software
- **Team ID-** 93881
- **Team Name –** Feature Fabricators

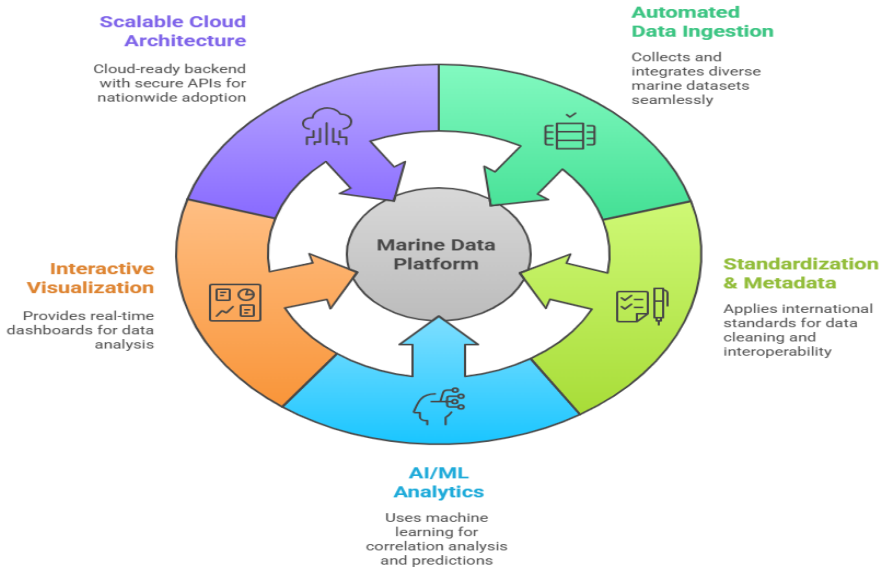
FEATURE FABRICATORS

IDEA TITLE

SMART INDIA HACKATHON 2025

❑ Proposed Solution

We propose to develop **OCCEANIQ : An AI-Driven Unified Data Platform** that integrates oceanographic, fisheries, and molecular biodiversity datasets into a single intelligent system.



1.Automated Data Ingestion – Collects and integrates structured, semi-structured, and unstructured marine datasets seamlessly.
2.Standardization & Metadata – Applies international standards for data cleaning, tagging, and interoperability.
3.AI/ML Analytics – Uses machine learning for correlation analysis, species identification, and ecosystem predictions.
4.Interactive Visualization – Provides real-time dashboards and modules for otolith, taxonomy, and eDNA analysis.
5.Scalable Cloud Architecture – Cloud-ready backend with modular microservices and secure APIs for nationwide adoption.

❑ How it Addresses the Problem

1.Breaks data silos by unifying heterogeneous marine datasets into a single platform.
2.Enables real-time monitoring and forecasting of ocean and biodiversity trends.
3.Simplifies access for scientists, policymakers, and conservationists.
4.Supports sustainable fisheries management with data-driven insights.
5.Improves decision-making for ecosystem health and Blue Economy initiatives.

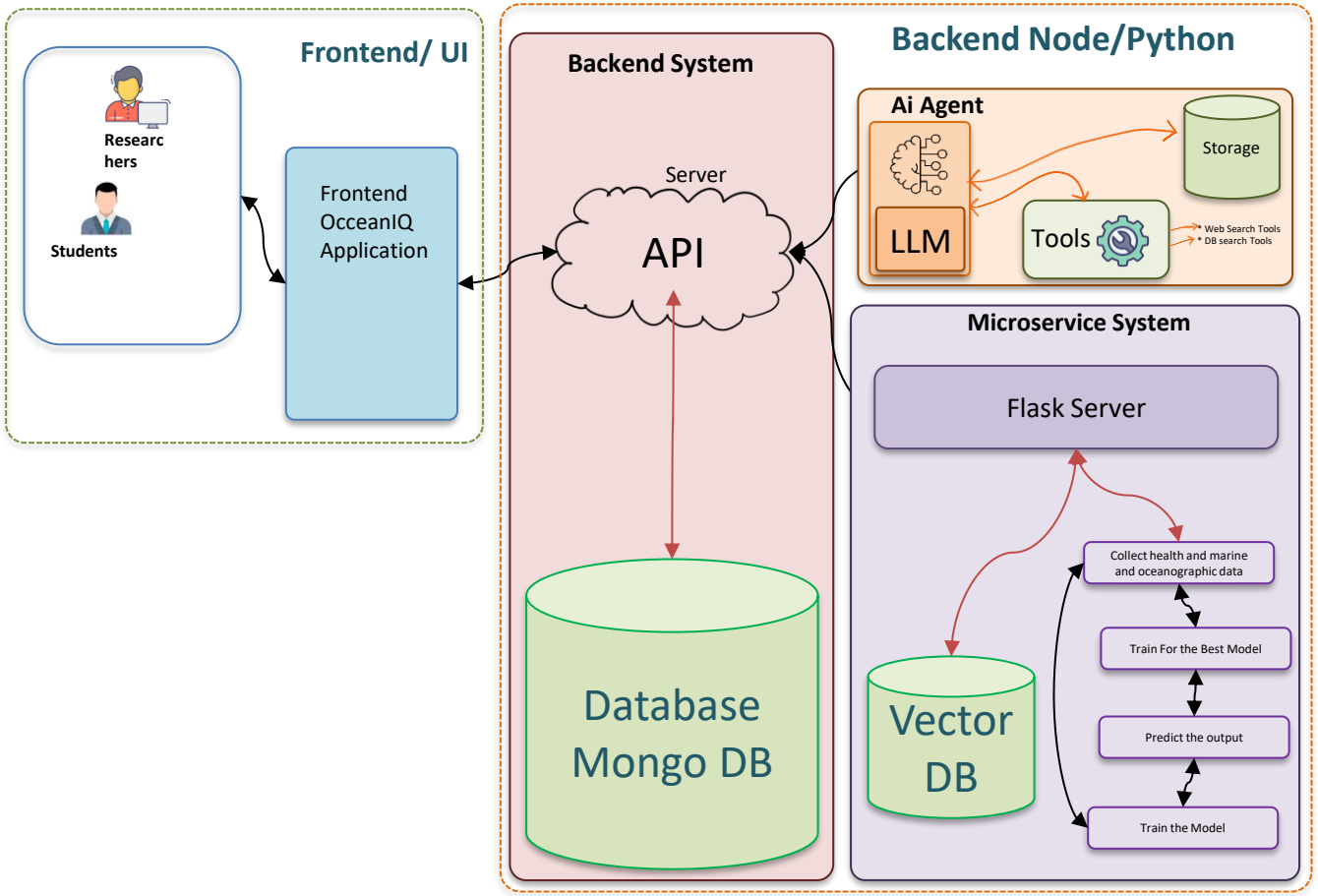
❑ Innovation and Uniqueness of the Solution



FEATURE FABRICATORS

TECHNICAL APPROACH

SMART INDIA HACKATHON 2025



01
Frontend
* React
* React icon
* Zustand

02
Backend
* Express
* mongoose
* Cors
* Bcrypt
* JsonwebToken

03
Styling
* Tailwind Css
* Lucid react

04
Database
* Mongo DB

05
Microservice
* Python
* Flsak
* Django

06
ML
* Scikit Learn
* TensorFlow

07
LLM
* Gemini-2.5-pro

08
Vector DB
* Chroma

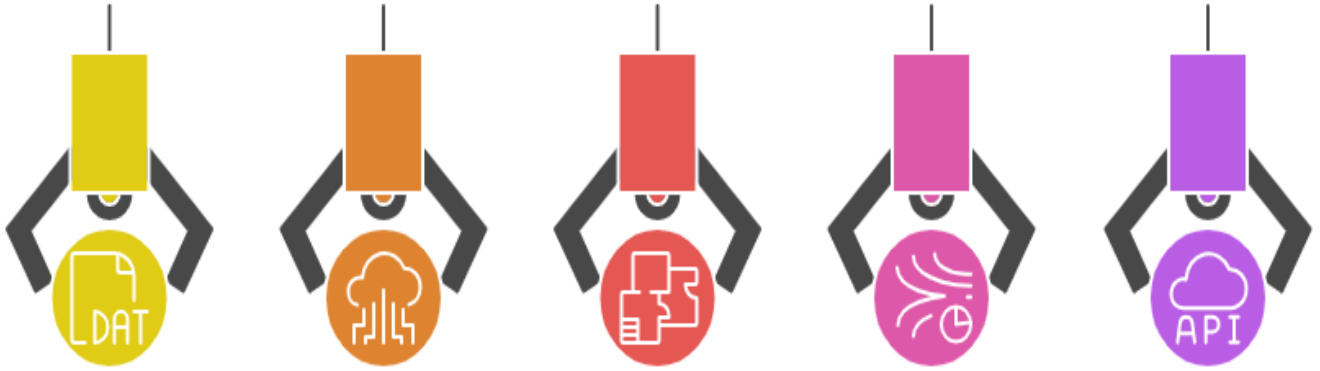
09
Deployment
* Docker container
* AWS

FEATURE FABRICATORS

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2025

❑ Feasibility:



Available Datasets

Datasets are available for pilot implementation.

Scalable Architecture

Cloud-ready scalable architecture is available for nationwide adoption.

Modular Design

Modular design ensures easy expansion.

AI/ML Proven

AI/ML is already proven in taxonomy & morphology analysis.

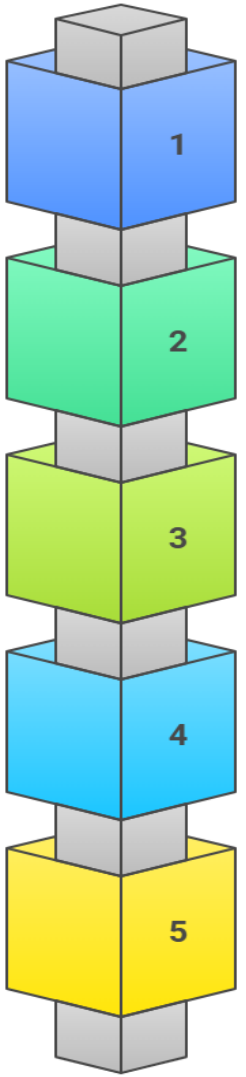
API Integration

APIs enable integration with international marine databases.

❑ Challenges & Risks:

- Handling large-scale heterogeneous data formats.
- Ensuring data quality and consistency.
- Limited labelled datasets for training AI/ML models.
- Cloud infrastructure cost and data security concerns.
- Resistance to adoption from traditional research practices.

❑ Strategies to Overcome:



Data Lakes

Establish schema-on-read data lakes for diverse datasets.

Data Preprocessing

Apply robust preprocessing and cleaning pipelines.

AI Model Training

Train AI models using transfer learning and synthetic datasets.

Security Protocols

Implement strong cloud security protocols and access control.

Training Workshops

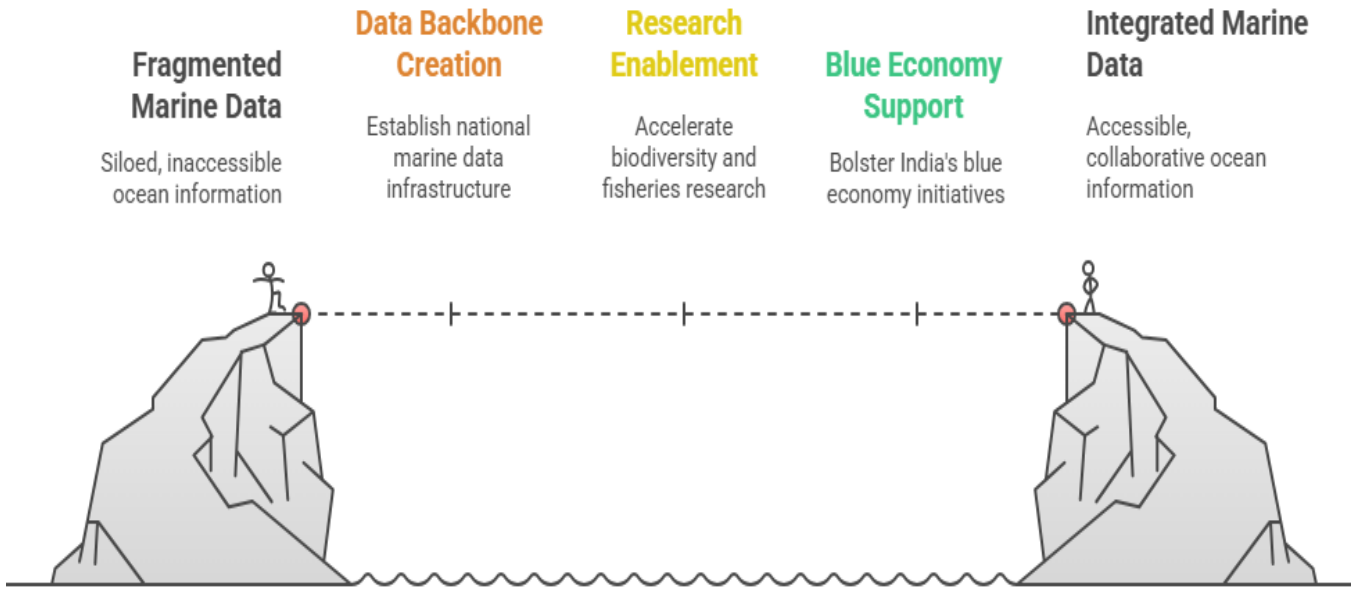
Conduct workshops to enable adoption by researchers.

FEATURE FABRICATORS

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2025

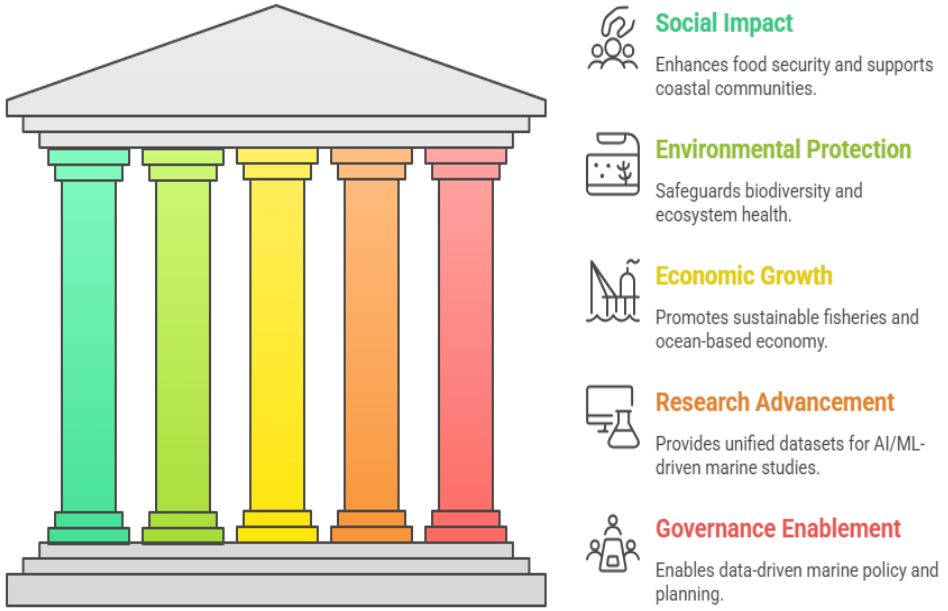
❑ Impact:



- Establishes India's **National Marine Data Backbone**.
- Enables faster biodiversity and fisheries research.
- Supports India's **Blue Economy** initiatives.
- Strengthens marine conservation and real-time climate monitoring.
- Enhances global collaborations in ocean and biodiversity research.

❑ Benefits:

- **Social:** Improves food security, supports coastal communities.
- **Environmental:** Protects biodiversity & ecosystem health.
- **Economic:** Boosts sustainable fisheries & ocean-based economy.
- **Research:** Provides unified datasets for AI/ML-driven marine studies.
- **Governance:** Enables data-driven marine policy and planning.



FEATURE FABRICATORS

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025

•**CMLRE, Ministry of Earth Sciences (Mo-ES), India**
Official website and marine biodiversity programs.
🔗 https://cmlre.gov.in
•**OBIS – Ocean Biodiversity Information System**
Global open-access database for marine species and ecosystem data.
🔗 https://obis.org
•**FAO Fisheries and Aquaculture Department**
Data standards, global statistics, and sustainable fisheries practices.
🔗 https://www.fao.org/fisheries/en/
•**ICES – International Council for the Exploration of the Sea**
Global standards for marine biology and oceanographic data integration.
🔗 https://ices.dk
•**NOAA – National Oceanic and Atmospheric Administration (USA)**
Research on ocean data, marine ecosystems, and climate analysis.
🔗 https://www.noaa.gov
•**Research Papers:**
•***Environmental DNA (eDNA) as a Tool for Marine Biodiversity Monitoring*** **– Nature Ecology & Evolution (2021)**
🔗 https://www.nature.com/articles/s41559-021-01498-6
•***Deep Learning for Fish Otolith Shape Classification*** **– Frontiers in Marine Science (2022)**
🔗 https://www.frontiersin.org/articles/10.3389/fmars.2022.890765/full
•*AI and Big Data in Oceanography: Towards an Integrated Marine Data Framework* – Marine Data Science Journal (2023)
🔗 https://www.sciencedirect.com/journal/marine-data-science
•**Data Standards & Frameworks:**
•**Darwin Core Standard (DwC):** https://dwc.tdwg.org/
•**OBIS Schema & Guidelines:** https://manual.obis.org/
•**INSPIRE Marine Data Specifications (EU):** https://inspire.ec.europa.eu/theme/ocean